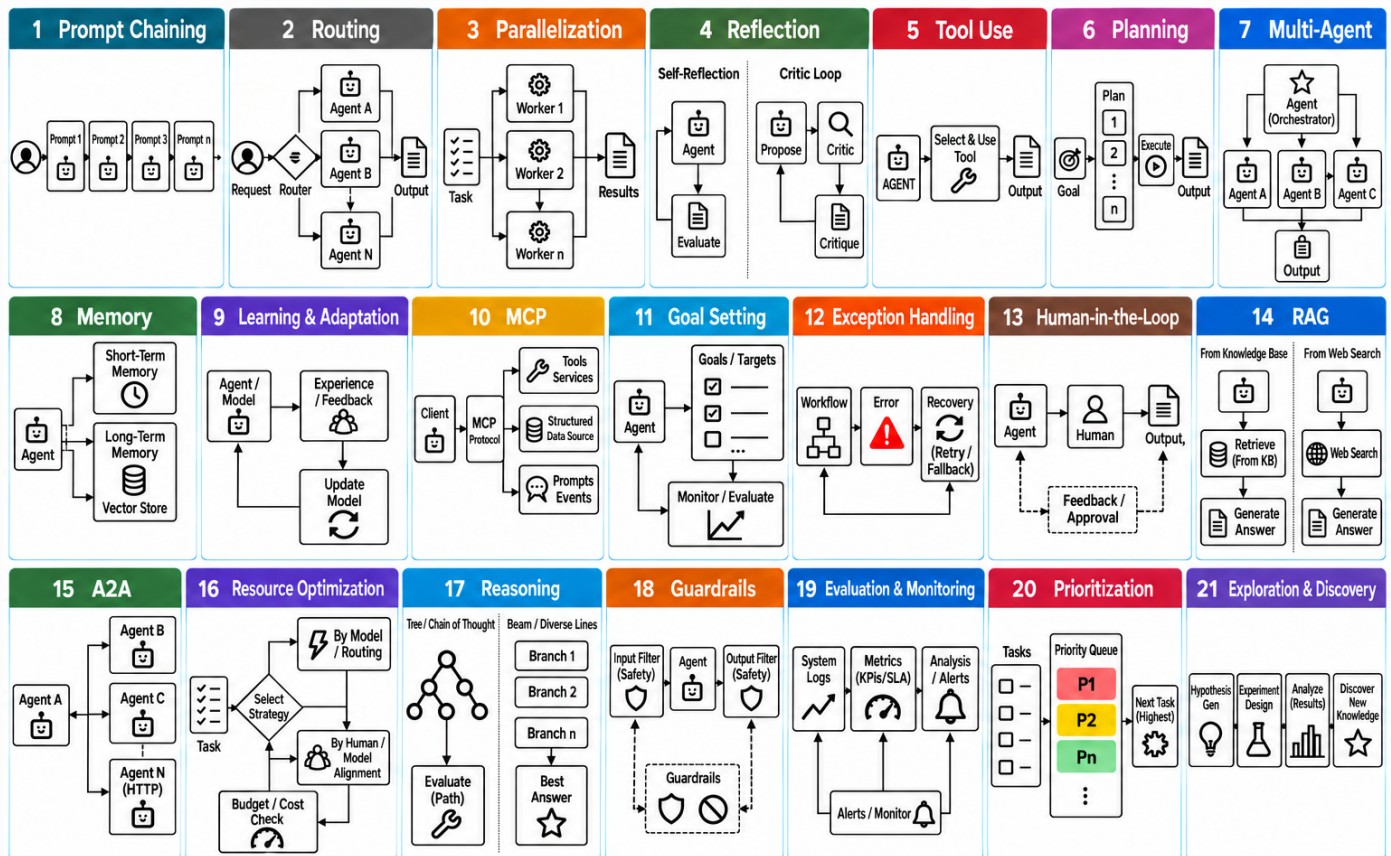


AGENTIC DESIGN PATTERNS

Main reference: [Agentic Design Patterns](#)

$$\text{Agentic AI} = \text{Model} + \underbrace{\text{Harness}}_{\text{Planning} + \text{Tools} + \text{State} + \text{Actions}}$$

Design Patterns: Below are 21 key design patterns for Agentic AI systems.



Pattern

Core Question

1. Prompt Chaining

How can a complex task be split into sequential steps?

2. Routing

Which workflow or capability should handle this input?

3. Parallelization

Which tasks can run concurrently?

4. Reflection	How can the agent review and improve its output?
5. Tool Use	How can the agent act through external tools and APIs?
6. Planning	How can a goal be turned into executable steps?
7. Multi-Agent Collaboration	How can specialized agents divide and coordinate work?
8. Memory Management	What context and past information should the agent retain?
9. Learning & Adaptation	How can experience and feedback improve future behavior?
10. Model Context Protocol (MCP)	How can agents access tools and resources through a standard interface?
11. Goal Setting & Monitoring	What defines success, and how is progress tracked?
12. Exception Handling & Recovery	How can the agent recover from failures or unexpected states?
13. Human-in-the-Loop (HITL)	When should a human review, approve, intervene, or take over?
14. Knowledge Retrieval (RAG)	How can the agent retrieve and use relevant external knowledge?
15. Inter-Agent Communication (A2A)	How can agents communicate and delegate through a standard protocol?
16. Resource-Aware Optimization	How should the agent trade off quality, cost, latency, and compute?
17. Reasoning Techniques	How can the agent reason, explore alternatives, and verify results?
18. Guardrails / Safety Patterns	What constraints must the agent follow, and how are they enforced?
19. Evaluation & Monitoring	How can agent quality be measured and degradation detected?
20. Prioritization	Which goal, task, or action should be handled first?
21. Exploration & Discovery	How can the agent discover new information, strategies, or possibilities?